Environmental Science & Technology

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Article

Rapid Mass Conversion for Environmental Microplastics of Diverse Shapes

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Cite This: Environ. Sci. Technol. 2024, 58, 10776−10785

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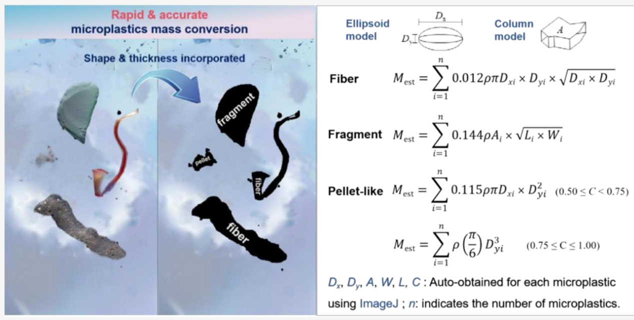
ABSTRACT: Rivers have been recognized as the primary conveyors of microplastics to the oceans, and seaward transport flux of riverine microplastics is an issue of global attention. However, there is a significant discrepancy in how microplastic concentration is expressed in field occurrence investigations (number concentration) and in mass flux (mass concentration). Of urgent need is to establish efficient conversion models to correlate these two important paradigms. Here, we first established an abundant environmental microplastic dataset and then employed a deep neural residual network (ResNet50) to successfully separate microplastics into fiber, fragment, and pellet shapes with 92.67% accuracy. We also used the circularity (C) parameter to represent the surface shape alteration of pellet-shaped microplastics, which always have a more uneven surface than other shapes. Furthermore, we added thickness information to two-dimensional images, which has been ignored by most prior research because labor-intensive processes were required. Eventually, a set of accurate models for microplastic mass conversion was developed, with absolute estimation *errors* of 7.1, 3.1, 0.2, and 0.9% for pellet ($0.50 \leq C < 0.75$), pellet ($0.75 \leq C \leq 1.00$), fiber, and fragment microplastics, respectively; environmental samples have validated that this set is significantly faster (saves ∼2 h/100 MPs) and less biased (7-fold lower estimation errors) compared to previous empirical models.



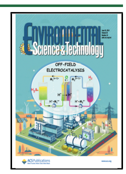
KEYWORDS: *environmental microplastics, shape autoclassification, thickness incorporation, mass conversion*

1. INTRODUCTION

The majority (∼80%) of plastic and microplastic pollution in the ocean originates from sources on land,[1] with the transport via rivers into the ocean being the primary route.[2−4] Microplastics are defined as plastic items with a size less than 5 mm, most of which are secondary formed ones as a result of the environmental weathering (e.g., wave breaking, ultraviolet irradiation, biological processes) on larger pieces of polymers.[5,6] As an emerging environmental contaminant, microplastics have garnered significant attention for their potential adverse impacts on the aquatic ecosystem; however, their pollution status and transport behavior need to be further investigated.[7−9]

Several global-scale model studies have examined the flux of microplastics into the ocean, but Weiss et al.[10] found that these studies have often overestimated the amount of microplastics entering the oceans from rivers. Several studies have used ellipsoidal models to convert the microplastic number to mass concentrations and added particle depth, but shape-separated conversion models have received little research yet. In the field investigations, the microplastic pollution status is always reported in number concentrations, e.g., particles/L and items/m^3,[11−13] because microscopic examination and polymer identification often process microplastics individually[14] and the small mass of microplastics makes it difficult to weigh. In order to convert to mass concentrations, previous studies usually equated light fiber-shaped microplastics with dense fragment-shaped ones (3.0 or 0.96 mg per item), and the error of this mass conversion alone may lead to an overestimation by 2−3 orders of magnitude.[3,10,15] Although Weiss et al. have largely improved the model by correcting the fiber-shaped microplastics with a median mass of 0.75 μg,[10] the conversion of concentrations from number to mass concentration is still very challenging. However, recent studies have addressed the conversion of number concentrations to mass concentrations by aligning the sizes and shapes of microplastics and incorporating depth.[16−18] In this study, we aimed to investigate if accurate microplastic mass conversion can also be obtained based on shape classification.

Received: January 28, 2024
Revised: May 27, 2024
Accepted: May 28, 2024
Published: June 5, 2024



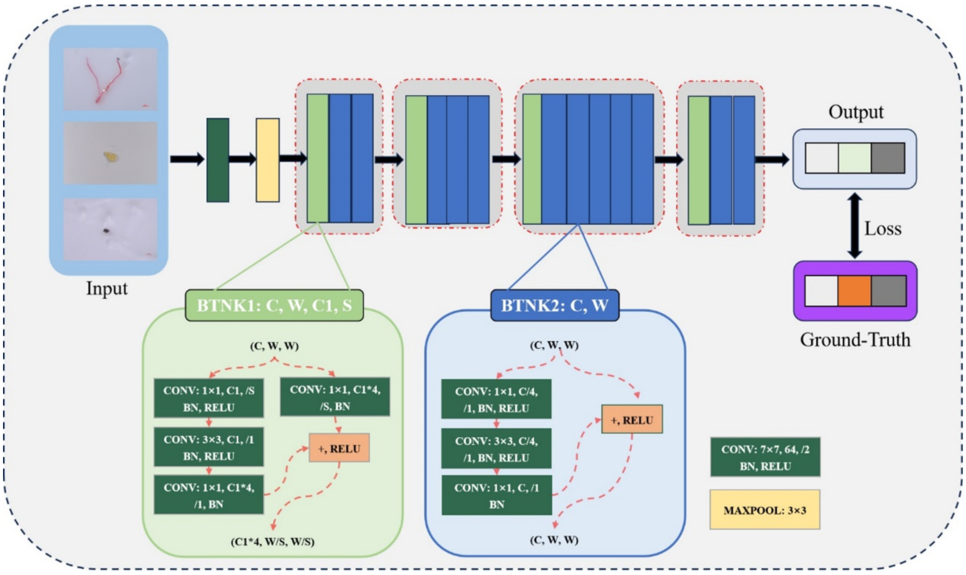


Figure 1. Architecture of ResNet50 utilized for microplastic shape classifying. The output of our model is a one-hot encoding of the corresponding label, which is used to calculate the loss value.

Accurate mass conversion of environmental microplastics requires the identification and classification of plastics with varying degrees of irregularity and different shapes, according to the parameterized probability density function (PDF) for more than 60,000 environmental microplastics.[17] For instance, the fiber-shaped microplastics are usually the dominant type, with 63−94% in the freshwater and marine water.[19−21] Besides, fragment- and pellet-shaped microplastics are the other two main shapes of microplastics in the aquatic environment.[22−26] The residual network (ResNet) is renowned as a highly effective end-to-end framework for image recognition.[27] ResNet and its variants have been widely utilized in high-level computer vision tasks such as semantic segmentation, image classification, and object tracking. Given the exceptional performance of the residual network in image recognition, it is very suitable to classify microplastic shapes; although previous research has used this model to separate microplastics, they used commercially available but not environmentally weathered irregular microplastics, which decreased its applicability in real scenarios.[28]

Upon the categorization of microplastic shapes, the development of mass conversion models encounters the extra obstacle of efficiently acquiring thickness information. In the past, various mass conversion models have been created for microplastics with specific shapes. For instance, prior studies have proposed employing cylinder models to estimate the mass of fibrous microplastics,[29,30] applying 40% porosity during the mass estimation according to Simon et al.'s research.[30] Additionally, empirical estimation of the mass of microplastics from micro-Fourier transform infrared (μ-FTIR) or FTIR data was also conducted by assuming an ellipsoidal shape,[31,32] whereas overestimation may occur for larger particles and increased particle numbers.[31] There are also shape-independent models for microplastic mass conversion. The ellipsoid models are suggested for mass conversion for all shapes of microplastics,[18,33] whereas the estimation accuracy of these models has seldom been clarified. Accurate mass calculations relied greatly on accurate thickness information. Nevertheless, the majority of existing empirical models has disregarded the influence of the thickness factor. In previous empirical models, thickness information was consistently missing because single-projection techniques were used, which can acquire only two-dimensional (2D) information. The acquisition of thickness necessitates the utilization of multiprojection analysis and/or roughness, which, despite being essential, can be quite laborious due to the requirement of multiple images from various projection directions.[16,34]

Thus, this study aims to incorporate shape classification and thickness information to create fast and efficient microplastic mass conversion models without additional analysis. First, we will automate microplastic shape sorting with artificial intelligence and then create geometric models that incorporate shape and thickness information. After comparing the estimation accuracy and time consumption of these models, we will find the most accurate and efficient ones and finally achieve the goal of giving out microplastic mass values without extra steps after field investigation.

2. MATERIALS AND METHODS

2.1. Dataset Preparation and Image Acquisition of Different-Shaped Microplastics. The single-projection analysis is considered a two-dimensional technique that is commonly used in field investigation protocols for studying microplastic pollution. We took microplastic images under the bright field of a Zeiss microscope equipped with a 9.0 million-pixel AxioCam digital camera. Sample set 1 included environmental microplastics (pellet, fiber, fragment), which were collected from our previous field investigation along the coastline of Chongming Island, China. As the number and irregular diversity of pellet-shaped microplastics were limited in the real environment, we also supplemented some commercial irregular pellet-shaped microplastics (including polypropylene, polyethylene, and polystyrene) from Sigma-Aldrich (Germany) into the sample set. The polymer compositions of these microplastics were identified using μ-FTIR (Thermo Scientific, Nicolet iS5N), and the degrees of polymer composition matching were all ≥80%.

The size range of microplastics was from 50 to 5000 μm (50−500 and 500−5000 μm) for the three shaped microplastics. The procedures for separating microplastics utilized a

https://doi.org/10.1021/acs.est.4c01031
Environ. Sci. Technol. 2024, 58, 10776−10785

series of stainless-steel sieves arranged in a cascade (Yinghua Hardware Instrument, China). Specifically, to obtain abundant data for the investigation of the surface morphology of pellet-shaped particles, we further divided them into additional size fractions (50−100, 100−200, 200−300, 300−400, 400−500, 500−600, 600−700, 700−800, 800−900, 900−5000). All of the weighed samples were stored in 1.5 mL centrifuge tubes. The weighing papers were checked for residual particles; if any remained, then, the numbers were added to the results. We injected 1000 μL of 100% ethanol (5 μm MCE membrane-filtered) into each tube and shook the tube bottom to top 10 times to disperse the particles evenly. Then, 1000 μL solutions of microplastics were immediately sucked out onto clean glass slides. The inner and outer walls of the pipet tips and tubes were washed with ethanol twice to rinse down any adhered ones. After the ethanol evaporated to dryness, we counted all of the microplastics on slides under the microscope. Single-projection photographs of fiber-, fragment-, and pellet-shaped microplastics were taken after positioning and focusing on a computer screen. Then, their high-contrast black-and-white images were obtained with microplastics as white silhouettes on a black background (Adobe Photoshop, version 22.0.0). The area, perimeter, fit ellipse, and Feret's diameter were obtained according to Merkus et al.,[35] and these data can be further used to enrich the PDFs of environmental microplastics' characteristics.[17]

To prepare samples for single projection, for each sample, their accurate mass was weighed using a precision balance (Sartorius BSA124S, Germany; accuracy 1.0×10^{-5} g), and the actual microplastic numbers were counted. In total, 83,702 microplastics were counted under the microscope (detailed numbers for each microplastic morphology are shown in Table S1).

2.2. Artificial Intelligence-Based Microplastic Shape Classification. We leveraged and modified the image classification residue network ResNet50 to classify the microplastic shapes. The structure of the ResNet50 architecture (Figure 1) illustrated the main workflow of our model, which followed an end-to-end framework. The neural network took microplastic images as inputs and generated predicted labels as outputs. To obtain the ground truth of microplastic shapes, we initially calculated the circularity. This allowed us to distinguish fibers (with circularity values ranging from 0.0 to 0.3) as well as some pellets (0.5−1.0) and fragments (0.3−0.9). Additionally, the width-to-length ratios for pellets (0.60−1.00) are typically greater than those for fragments (0.10−0.80). Ultimately, if there are any remaining particles that cannot be differentiated, we seek the expertise of three microplastic researchers to ascertain their shapes.

We first constructed a training dataset by inputting the above 2D contrast images of microplastics with different shapes taken under the microscope. These contrast images were first manually separated into different shapes, namely, pellet-, fiber-, and fragment-shaped microplastics. The dataset for particle shape machine learning consists of randomly selected 3024 real-world images of microplastics. These shapes are categorized into three labels: fiber, fragment, and pellet. Each image was input with a size of 512 × 512 × 3 during the preprocessing stage. We randomly selected 20% of the total images for the validation dataset, while the remaining images were used for the training dataset. As microplastic shape classifying is involved in a multiple-label prediction task, therefore, we adopted the cross-entropy loss as our loss function, which is expressed as

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c}^{K} y_{ic}\log(p_{ic}(x_i)) \tag{1}$$

where N is the number of images in a batch, i is the index of images in the current batch, c is the label index, K is the number of all labels, p_{ic} is the output of the network, and x_i is the input sample in a batch.

To evaluate the performance of our applied model, we adopted two metrics, including Accuracy and F_1-score, which are defined as

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{2}$$

$$F_1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{3}$$

where TP and TN are the amounts of true positive and true negative labels, respectively, and FP and FN are the corresponding amounts of false positive and negative labels.

We implemented the shape classification on a computer with Ubuntu 20.04.3, 2 Intel(R) Xeon(R) CPUs, and 4 GTX 3090 GPUs. The batch size was set as 8, and the learning rate was assigned to 1e−4. Our applied network was optimized by the Adam optimizer with 50 epochs in the Pytorch framework. The codes and data have been wrapped up and can be found at https://github.com/qihuiqing.

2.3. Multiprojection-Based Thickness Conversion. We performed the multiprojection analysis to obtain empirical thickness conversion equations. The multiprojection analysis was selected because it is a three-dimensional (3D) method and particles can be microscopically imaged from six orientations.[34] As the geometric models used in this study include ellipsoid, cube, and cylinder ones, the irregular particles were the most suitable shape for multiprojection analysis to establish the relationship between the thickness and other dimensional parameters.[30]

Microplastics were viewed in completely random orientations during the capturing of multiprojection photographs.[18,34] As the microplastics tended to stabilize on their maximum plane, here, we used vinyl glue and tweezers to randomly change the angle of each item and image them under the microscope from six different directions. Five particles were randomly selected within each replicate of different sizes among 50−5000 μm. Three maximum area and three minimum area projection photographs were all captured for 630 microplastics in total, and 3780 photographs were further processed with object selection and filling to obtain black-and-white images.

The thickness equations of $T_1 = W^2/L$ and $D_{z_1} = D_y^2/D_x$ for microplastics (Table S2) were obtained based on the relationships with length, width, and diameter.[18,30] The other four thicknesses, including $T_2 = k_1 W$, $T_3 = k_2 L$, $D_{z_2} = k_3 D_y$, and $D_{z_3} = k_4 D_x$, were created based on k_{1-4} analyzed from all multiprojection images. These geometric relations are applicable to all microplastics in the solid form of microplastics ranging from 50−5000 μm, and then, only the single-projection analysis is required during the application, but 3D information can be achieved meanwhile.

https://doi.org/10.1021/acs.est.4c01031
Environ. Sci. Technol. 2024, 58, 10776−10785

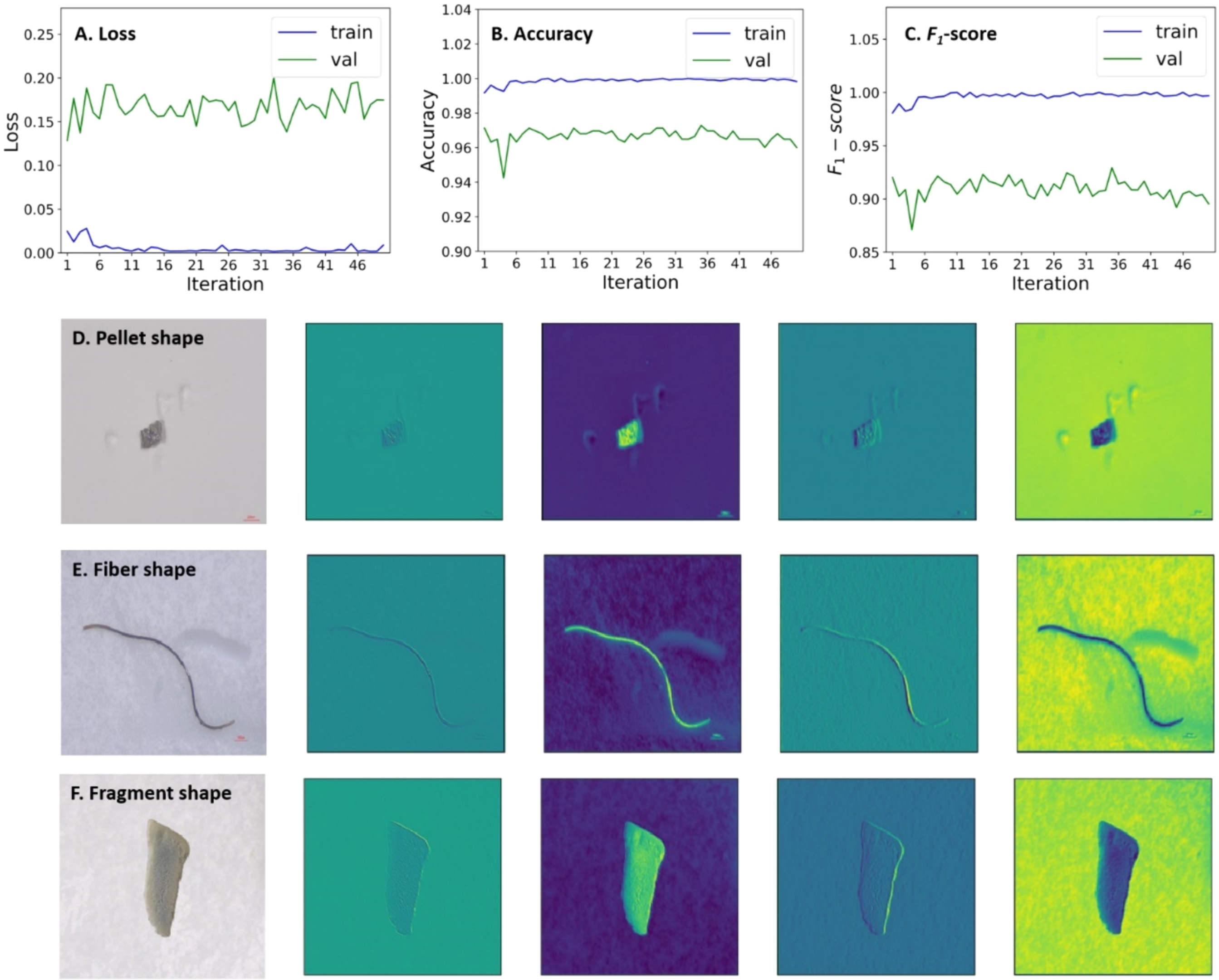


Figure 2. Detailed trends of the training process and the intermediate feature maps of three shapes of microplastics. (A) Training loss value at the beginning; (B) train and validation accuracy rates; (C) F1 scores of the train and validation; and the trained network examples of (D) pellet-shaped, (E) fiber-shaped, and (F) fragment-shaped microplastics. The first columns of (D−F) are original input images, while the second to the fifth columns are feature maps from the trained network. Val: validation.

2.4. Different-Shaped Microplastic Mass Conversion Model Establishment. To estimate the mass (M_{est}) of pellet-shaped microplastics from single-projection analysis, 24 models of spheres, ellipsoids, cylinders, and cubes combined with the characteristic parameters were established (Table S3). Herein, models 1−7 are simple spheres using different size parameters, and models 8−24 are different geometric models that incorporated the thickness information derived from 3D multiprojection analysis. Like particles, based on the thickness approximation, geometric mass conversion models were also established for fiber-shaped microplastics (ellipsoidal models 35−38 and the cylindrical models 39−42; Table S4) and fragment-shaped microplastics (ellipsoidal models 43−46 and cylindrical models 47−50; Table S5). The governing equations of spherical, ellipsoidal, cylindrical, cubic, and column models used during the conversion process are given in Text S1.

Specifically, the pellet-shaped microplastics have very diverse surface shape alterations, and thus, we also analyzed four surface shape descriptor parameters (circularity, aspect ratio, roundness, and solidity). Circularity (C) is a parameter describing the degree of roundness of a particle, with consideration of the perimeter smoothness.[36] The aspect ratio (AR) reflects the overall elongation of a particle.[37] Roundness reflects how closely a pellet approaches a perfect circle.[38] Solidity reflects the degree to which the area and convex hull area approach each other.[36] Besides, empirical models 25−30 (3D) and 31−34 (2D) reported in the literature are also included for comparison.[30,34,39−44]

2.5. Mass Conversion Model Verification and Application. A mixture of environmental microplastic pellets, fibers, and fragments (sample set 2) along the coastal lines of Chongming Island (Shanghai, China; N 31°33′, E 121° 56′) was collected to validate our models for mass conversion, and the accuracy was compared with empirical models. Estimation *error* (%) was calculated for each model as follows

$$\text{estimation error (\%)} = \frac{M_{est} - M_{mea}}{M_{mea}} \times 100\% \quad (4)$$

where M_{est} is the estimated mass, which was calculated based on models for each sample, and M_{mea} is the measured mass,

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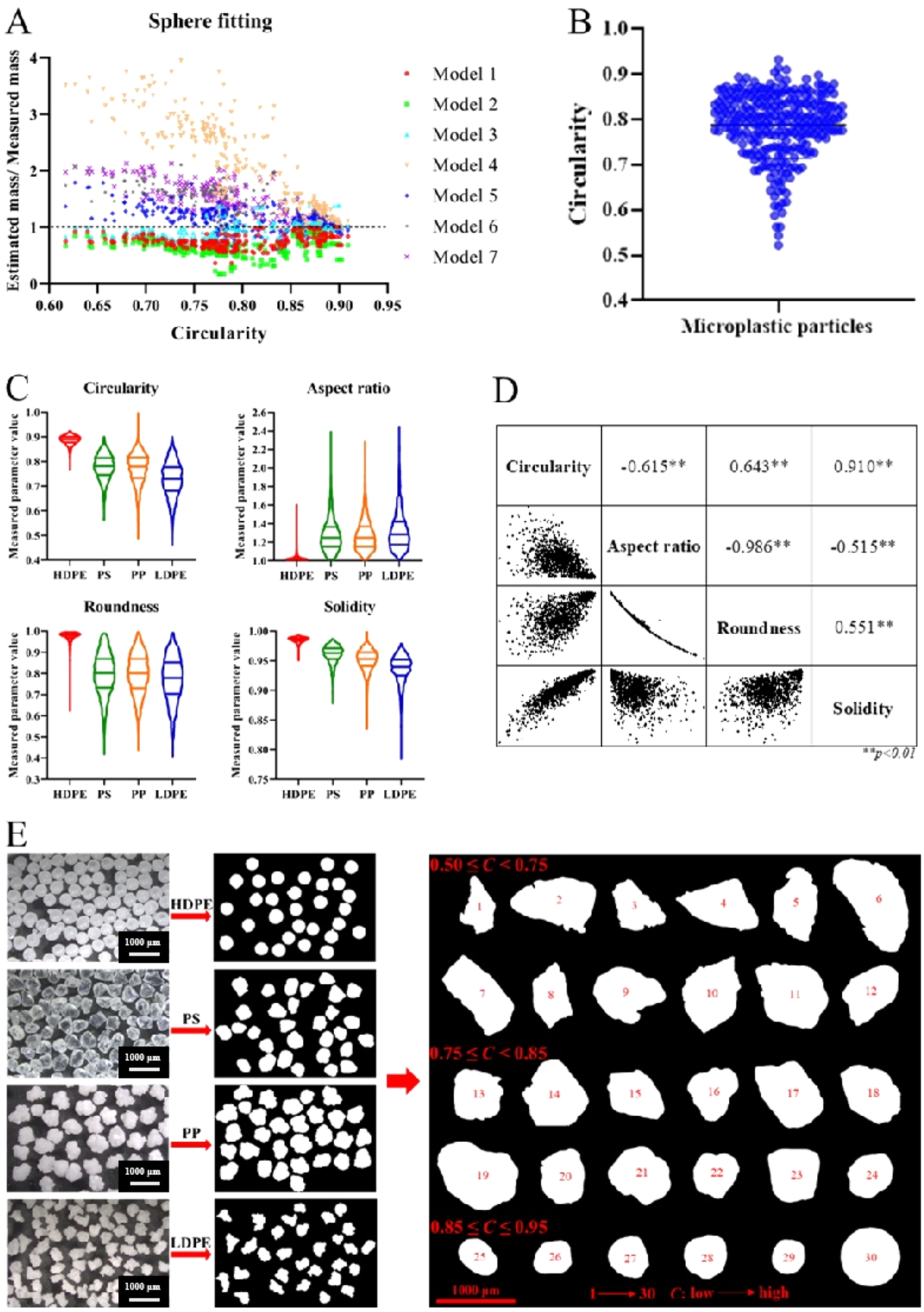

Figure 3. Surface shape descriptor analysis for pellet-shaped microplastics. (A) Relationship between M_{est}/M_{mea} calculated by models 1−7 and circularity. (B) Circularity range of microplastics collected from the natural environment along the coastline of Chongming Island, China. (C) Value ranges of four shape factors of the four types of microplastics. The three lines on violin plots are the upper quartile, median, and lower quartile. (D) Scatter plots and Spearman correlation coefficients between the pellet shape factors, and ** indicates a strong linear correlation between them ($p < 0.01$). (E) Single-projection images of representative morphologies of pellet-shaped microplastics with circularity values varying from 0.50 to 0.95. Note: to include various shape values, this study used four pellet-shaped microplastics: high-density polyethylene (HDPE), polystyrene (PS), polypropylene (PP), and low-density polyethylene (LDPE) with decreasing circularity.

which was obtained by dividing the total mass by the item number of each sample.

2.6. Quality Control and Assurance. All the above processes were completed in a biosafety cabinet to reduce the likelihood of airborne contamination,[45] and the sieves were washed with Milli-Q water to remove any potential particles that may have adhered, wrapped in foil, and dried in an electrothermal constant-temperature incubator (HPX-9052MBE, Shanghai, China). We performed microplastic-free sieving for 30 min in the first, middle, and later stages of the sieving process, and no microplastics >50 μm were found under the stereomicroscope. Cotton white lab coats were worn all the time during the sample analysis. Parallel procedural blank controls ($n = 3$) were conducted throughout the whole process of the measurement, where none of the microplastics were observed throughout the process.

https://doi.org/10.1021/acs.est.4c01031
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2.7. Data Analysis. The results were analyzed and plotted using SPSS (version 19.0, IBM), GraphPad Prism (version 8.0, GraphPad), and OriginPro (version 9.0, OriginLab). All data first passed the Shapiro−Wilk normality test, and then, the t test was applied for two groups, and analysis of variance (ANOVA) analysis with the LSD post hoc test was applied for three or more groups. The significant difference level was set as $p < 0.05$.

3. RESULTS AND DISCUSSION

3.1. Artificial Intelligence-Assisted Automatic Microplastic Shape Classification. We mainly present these intermediate results of training and the final numerical recognition results. The trend of loss, accuracy, and F_1 score are shown in Figure 2. It is obvious that the training loss value is almost convergent to 0 in Figure 2A. In the plot of Figure 2B, the training and validation accuracy rates are both higher than 94%. Figure 2C shows F_1 scores of the training and validation, which are higher than 0.85. We save the best model on the whole validation dataset for testing.

The convolutional kernels in our trained network demonstrated their effectiveness in extracting shape features and background information from the input images (Figure S1). The network can also generate intermediate feature maps corresponding to the pellet, fiber, and fragment classes (Figure 2D−F). These feature maps clearly exhibit distinctive and significant features that enable the discrimination of microplastic shapes. The combined findings demonstrate the strong recognition capabilities of our model, when it comes to identifying microplastic shapes. During the validation, results showed that our model accurately predicted all instances labeled as 'fiber' in the test dataset. For the 'fragment' class, our model successfully distinguished 51 out of 63 inputs. Similarly, for the 'pellet' class, our model achieved 56 correct predictions out of 63 samples.

Overall, we utilized the ResNet50 architecture for microplastic shape classification, which yielded a remarkable performance. Prior research has utilized this model for the purpose of segregating microplastics. However, they employed commercial spherical microplastics that have not been subjected to environmental weathering.[28] The proposed model successfully distinguished various microplastic shapes, facilitating subsequent calculations of mass concentration for microplastics. While our model may encounter challenges in predicting complex background scenes and transparent plastics, it still achieved an impressive accuracy rate of 92.67%, making it highly practical and effective.

In this study, microplastic particles were projected and identified one by one, and thus, the particle overlap situation did not occur. In future studies, we may use image scanning methods for a number of particles' projection together, and then, the adoption of models (i.e., the AISFormer model) to detect overlaps and/or impurities and segment the particle edges clearly is of vital importance.

3.2. Shape Alteration Description for Pellet-Shaped Microplastics. In comparison to fiber- and fragment-shaped ones, pellet-shaped microplastics are relatively more difficult to clearly depict due to their surface shape alterations. Environmental pellet-shaped microplastics often have convex hulls or elongated regions, and the estimation *errors* will be highly influenced by the surface irregularity.[46,47] For instance, with the decreasing circularity, the estimation *errors* for perfect sphere models 1−7 were increasing (Figures 3A and S2). Therefore, to create accurate models for pellet-shaped microplastics, it is essential to take shape description parameters into account.

In general, the surface shape descriptors indicated that most pellet-shaped microplastics were far from standard spheres[46] and had varying degrees of elongation and convex forms. According to our analysis, the circularity values were within the range of 0.5−1.0 for environmental microplastic pellets (Figure 3B), and the values of aspect ratio, roundness, and solidity were within the range of 1.0−2.5, 0.4−1.0, and 0.7−1.0, respectively (Figure 3C). We found that the four shape descriptors were highly or moderately correlated with each other according to the Spearman correlation results. Particularly, the circularity values were highly correlated with the solidity values ($r = 0.910$, $p < 0.01$) and significantly correlated with the aspect ratio ($r = -0.615$, $p < 0.01$) and roundness values ($r = 0.643$, $p < 0.01$) (Figure 3D). Additionally, previous research indicates that there is a strong correlation between the 2D circularity and the 3D sphericity of a pellet, provided that the pellet is not flat.[39,48] Since the model can perform efficient shape classification, these studies strongly support circularity as a good description of pellet shape alterations. Thus, circularity was chosen as a representative parameter for surface shape description of pellet-shaped microplastics (Figure 3E).

3.3. Thickness Derivation from Multiprojection Analysis. With the 3D multiprojection analysis, we obtained six metric equations for microplastic thickness calculation (Table S2). We calculated that the T_m/W_m and T_m/L_m ratios of irregular microplastics were quite constant at 0.67 ± 0.13 and 0.54 ± 0.14, respectively. Thus, for column and cube models, three thicknesses of $T_1 = W^2/L$, $T_2 = 0.67W$, and $T_3 = 0.54L$ were derived. Similarly, for ellipsoid models, the z-axis diameters of the fitting ellipsoids were also derived: $D_{z1} = D_y^2/D_x$, $D_{z2} = 0.69D_y$, and $D_{z3} = 0.55D_x$. The above results were obtained from multiprojection analysis of microplastics and were very suitable to describe the environmental microplastics.[18,33] While it is also possible to acquire thickness information through roughness analysis,[16] we used the multiprojection method due to its applicability to various models beyond the ellipsoidal model. In particular, a very similar T_m/W_m ratio of 0.67 ± 0.22 reported previously for microplastics at wastewater treatment plants[30] further verified our results.

3.4. Microplastic Mass Estimation Based on Thickness-Incorporated Models. The 3D multiprojection method can fully reflect the characteristics of microplastics, but it is cumbersome as multiple images from different projection directions are required to be taken.[34,49] Therefore, a faster and more efficient forecasting method is urgently needed, and a renovated thickness-incorporated single-projection method is a promising strategy for addressing this issue and was developed in this study.

For pellet-shaped microplastic mass conversion models, 16 renovated thickness-incorporated projection models were used (models 8−24; Table S3). Several renovated thickness-incorporated models achieved good mass estimation accuracy (Figure S2). The changes in circularity, a parameter describing the degree of roundness, can strongly affect the mass estimation *errors*. The estimation *errors* using model 13 for the circularity of $0.75 \leq C < 0.85$ and $0.85 \leq C < 0.95$ were only 2.0 and −3.1%, respectively, exhibiting the best accuracy (Figure 4A,B). For the circularity of $0.50 \leq C < 0.75$, model 13

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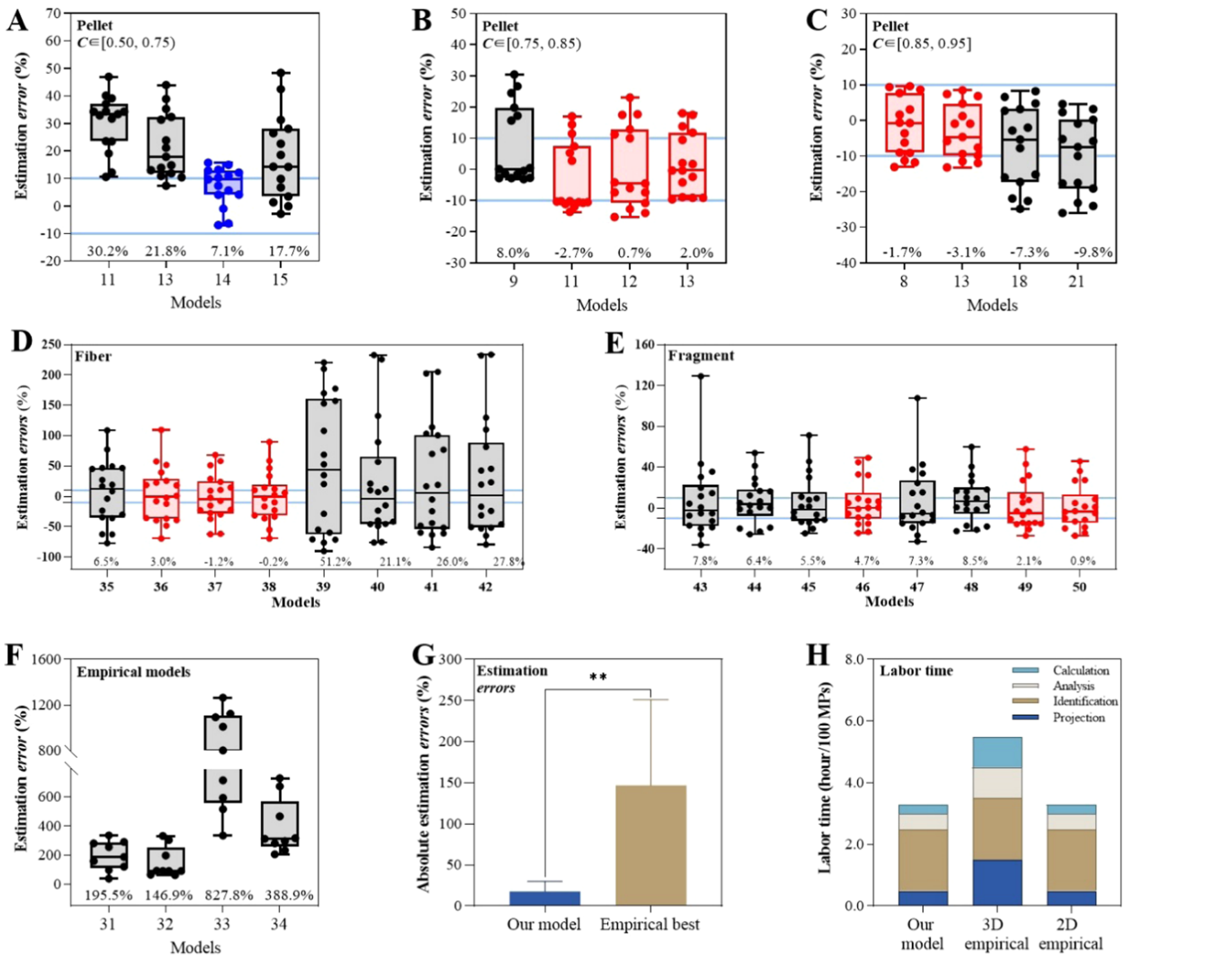


Figure 4. Estimation *errors* of environmental microplastics by using the models created in this study for pellet-shaped microplastics (A−C), fiber-shaped microplastics (D), fragment-shaped microplastics (E), 2D empirical models (F); and the comparison between our model set and empirical models on estimation *errors* (G) and labor time (H). *Projection*: particles image projection; *Identification*: polymer composition identification; *Analysis*: size parameters analysis with ImageJ; and *Calculation*: mass estimation calculation. In each box plot, the ends of the bars represent the smallest and largest *errors* (%). The box height indicates the first and third quartiles and the horizontal line in the box is the median (second quartile) of the measurement. The points in box plots are all *error* (%) values. The red box plots highlight the models with mean estimation *error* (%) within ±5%. The blue box plot highlights the model with the lowest mean mass estimation *error* (%) within ±10%. C is the circularity value for pellet-shaped microplastics. Numbers at the bottom of box plots are the mean of all *error* (%) values. ** indicates that the corresponding models showed significant differences with p values <0.01.

was no more suitable as the estimation *error* reached 21.8%, whereas model 14 was the best-performing model with an estimation *error* of −7.1% but a bit higher than the 5% threshold (Figure 4C and Table S6). Since model 13 was also applicable to pellets of $0.95 < C \leq 1.00$, we ultimately divided pellet-shaped microplastics into two categories by circularity ($0.50 \leq C < 0.75$ and $0.75 \leq C \leq 1.00$; Table 1).

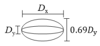
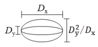
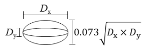
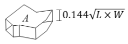
In contrast, only limited empirical pellet-shaped models exhibited acceptable fitting goodness, and empirical 3D models perform better than their 2D counterparts. Among the empirical 3D models, models 25 and 30 had R^2 values around 0.95, largely because the strong relationship between volume and area$^{1.5}$ was considered in these equations.[34,40] Besides, only one of the 2D empirical models (model 32)[30] exhibited an acceptable estimation result with the goodness of fit above 0.9 (Figure S2). Further, relatively low accuracy was obtained for most empirical models, except for models 28 and 32, which reached low estimation *errors* of −1.3 ± 11.0% and 4.0 ± 6.7% for microplastic pellets of $0.50 \leq C < 0.75$, respectively.

For fiber-shaped microplastic mass conversion models, the analysis results showed that the ellipsoidal models achieved better mass estimation accuracy than the cylindrical models, with mean errors of less than ±10% (Table S7). We marked the models with mean errors within ±5% in red, namely model 36, model 37, and model 38, with a mean ± SEM of 3.0 ± 10.6, −1.2 ± 8.9, and −0.2 ± 9.5%, respectively; and model 38 was shown to have the best prediction performance (Figure 4D). However, the mean of errors generated by the cylindrical models here exceeds 20%, causing obvious overestimation. This may be due to the fact that the size parameters L and W of the cylinder are related to the Feret diameter. When measuring fibrous microplastics, the Feret size often exaggerates the actual size of the projected contour,[50] resulting in significant errors in volume and mass model predictions.

For fragment-shaped microplastic mass conversion models, model 46 had the smallest mean error of 4.7 ± 5.2% within all the ellipsoidal models; moreover, the cylindrical models 49 and 50 achieved higher mass estimation accuracy, with errors of 2.1 ± 5.6 and 0.9 ± 5.0%, respectively, and the column model 50

https://doi.org/10.1021/acs.est.4c01031
Environ. Sci. Technol. 2024, 58, 10776−10785

Table 1. Screened-Out Mass Conversion Model for Differently Shaped Microplastics

Shape [a]	Equation [bcd]	Model
Pellet	$M_{\text{est}} = 0.115\rho\pi D_x \times D_y^2$ $(0.50 \le C < 0.75)$	
	$M_{\text{est}} = \rho\left(\frac{\pi}{6}\right)D_y^3$ $(0.75 \le C \le 1.00)$	
Fiber	$M_{\text{est}} = 0.012\rho\pi D_x \times D_y \times \sqrt{D_x \times D_y}$	
Fragment	$M_{\text{est}} = 0.144\rho A \times \sqrt{L \times W}$	

[a] C is the circularity of microplastics; it can be directly obtained from ImageJ measurements (shape descriptors). [b] M_{est} is the average estimated mass of each microplastic. [c] D_x and D_y are the major and minor diameters of the particles, respectively; these two size parameters can be directly obtained from ImageJ measurements (fit ellipse). [d] L and W are the maximal and minimal Feret diameters of particle projection, respectively.

was shown to be the best-suited one for mass conversion of microplastic fragments, which can decrease 3.6% estimation *errors* more than the best-fit ellipsoidal model 46 for fragments (Figure 4E and Table S8).

3.5. Microplastic Mass Conversion Model Verification and Comparison. For further comparison among the screened-out models, we also used the environmental microplastic sample set 2 to verify the four screened-out models (Table 1), whose size distribution can be found in Figure S3. The mass estimation *errors* of the empirical models for different shapes of microplastics were very high (146.9–827.8%; Figure 4F and Table S9). Obviously, the estimation *error* (mean ± SEM of 18.0 ± 4.1%) of our model set for all shaped microplastics was much smaller than these empirical models significantly ($p < 0.01$; Figure 4G and Table S9).

Thus, for the previous 2D empirical models 31–34, they are less effective at predicting the mass of complex microplastics, and their *errors* were much higher (from 146.9 to 827.8%) than our model set (Table S9). Other kinds of empirical models, such as the 3D empirical Model 28, cannot be applied to compare as they need to obtain multiple projection images for each item before calculation, which is impractical and time-consuming during real investigations.

We also compared the estimated mass values using our model to the values obtained from pyrolysis-GC/MS for environmental samples. Primpke et al.[31] have provided valuable information on both the average dimensional parameters of microplastics and the measured mass values by pyrolysis-GC/MS. According to the sample images, we can see that the analyzed microplastics are fragment- or pellet-shaped. As most of the polypropylene microplastics in the environment exist in the shape of fragments, we selected the polypropylene microplastics from the sample Oldenburg 1308VF for mass conversion. We used the pyrolysis-GC/MS mass result as the reference value (0.0003173 g)[31] and compared it with the calculated value from our model (0.0003315 g). The estimation *error* between our model and the pyrolysis-GC/MS method is 4.5%. It suggests that our model performs excellently in converting microplastic fragment numbers to mass. However, due to the unavailability of 2D images and thus the shapes of all microplastics, we were unable to compare and analyze all the samples. In the future, more comparisons between the calculated values and pyrolysis-GC/MS-measured values for environmental samples are needed to verify the estimation accuracy of the models.

Moreover, the labor time required to perform image projection (Projection), polymer identification (Identification), size parameter analysis (Analysis), and mass estimation calculation (Calculation) was piled up for these screened-out models. Both our model set and 2D empirical models (i.e., Simon et al.'s model,[30] which has been widely applied in mass conversion studies[31,32]) require approximately 3 h to analyze 100 microplastics, while 3D empirical models require about 2 additional hours for the particles' 3D multiple projection (Figure 4H), which is not practical for actual environmental microplastic sample analysis.[34] Additionally, no extra time is required if researchers use our automatic shape classification and mass conversion model set (Figure S4) because the 2D image capture of environmental microplastics is a standard step for field investigation. In further research, our method can also integrate the image processing approach for meso- and macroplastic debris,[51] to quantify the riverine flux of all plastics to the oceans.

A robust method for converting the number, volume, and mass concentrations of environmental microplastics was recently proposed by researchers of Prof. Koelmans' group, by employing a continuous description of the morphotypes and a realistic representation of the "true" environmental microplastics.[18,33] While these exemplary studies were in line with the goal of this research, we noticed that ellipsoidal models might not consistently be the most effective models. Thus, we established specific models for pellet-, fiber-, and fragment-shaped microplastics separately. Additionally, it is possible to align and translate diverse size range datasets into a unified range according to Koelmans et al.[18] In contrast, the mass conversion for microplastics was performed directly on the basis of the actual size measurement without any translation in this study.

Collectively, our shape and thickness factor-incorporated models showed an excellent performance in terms of accuracy and efficiency. The established model set in this study can also be easily applied to other studies that provide single-particle data. Therefore, the present work established connections between investigations of microplastic pollution and studies of transport flux and/or effects. It is feasible to obtain precise transport flux for microplastics, and the equations can also be used to convert the number concentration to mass concentration for previous studies that have taken single-projected images of microplastics. In addition, a framework for future risk assessment can be developed based on this study, which offers a method for concentration alignment between actual environmental exposure and laboratory effect data.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.4c01031.

Equations for different geometric models (Text S1); mass and number values of microplastics (Table S1); thickness conversion formula based on multiprojection analysis (Table S2); detailed description and diagrams of models 1–50 (Tables S3, S4 and S5); mass estimation *errors* of pellet shaped, fiber-shaped, and fragment shaped microplastic mass models (Tables S6, S7, and S8); mass estimation errors of environment-mixed samples (Table S9); trained kernels of the first layer in the ResNet50 mode (Figure S1); comparisons of the estimated mass derived from 2D and 3D models (Figure S2); size distribution of environmental microplastics (Figure S3), and renovated processes for microplastic number and mass concentration measurements (Figure S4) (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Key Research and Development Program of China (2022YFC3105900) and the National Natural Science Foundation of China (Nos. 42077371, 42342057 to Q.C.). It is also supported by the Open Research Fund of State Key Laboratory of Estuarine and Coastal Research (SKLEC-KF202311 to L.S.).

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Environ. Sci. Technol. 2024, 58, 10776−10785